

what the bits mean

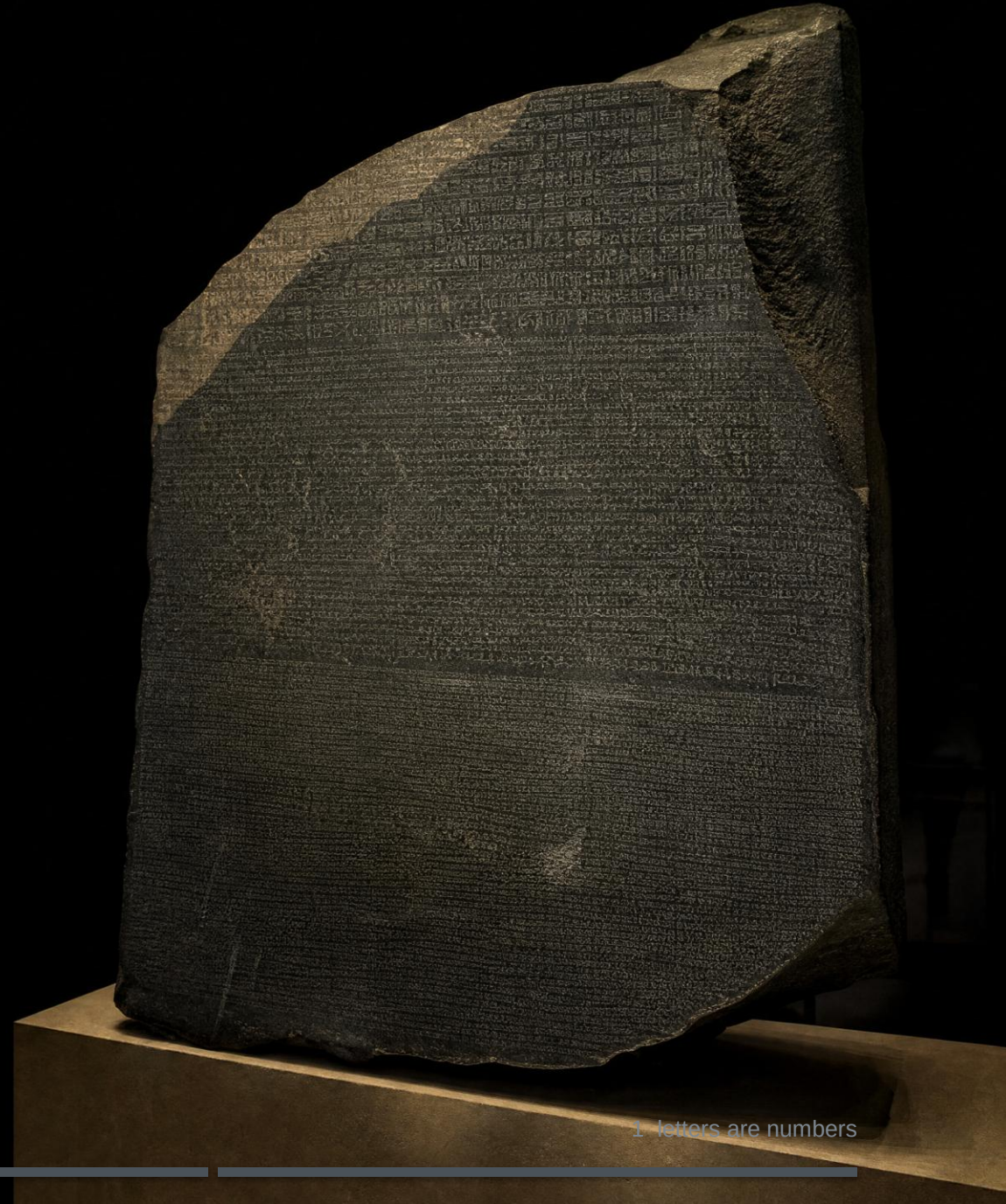
digitalisierung und programmierung · input, session 6

agenda

- 1 letters are numbers
- 2 how long is a character
- 3 the same bits, other meanings

letters are numbers

who still knows
what this means?



a code system maps characters to sequences of symbols.

agreed, not given by nature. that is why there are several of them.

ascii

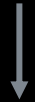
A



65

01000001

B



66

01000010

C



67

01000011

a



97

01100001

a letter is a number. the letter appears only when someone draws it again.

ascii: 128 characters

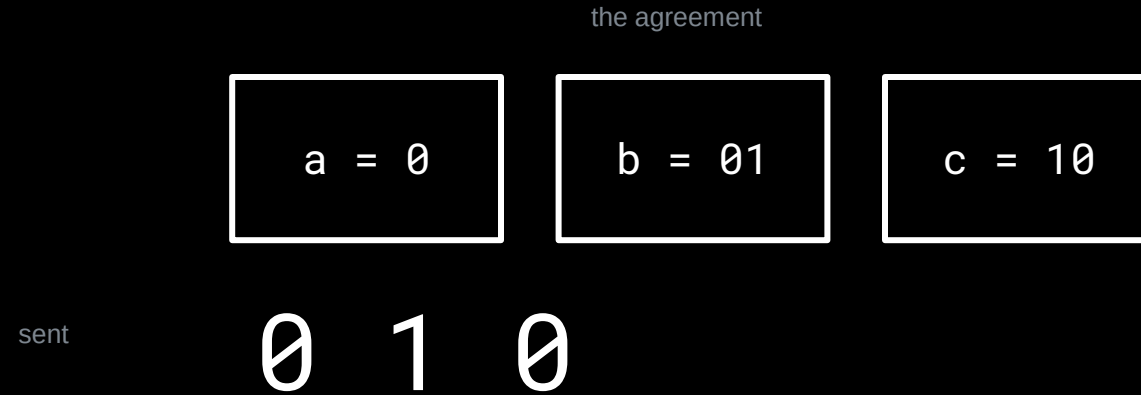
not enough for the world.

unicode gives every character a number,
over 100,000 of them.

utf-8 writes common ones in one byte,
rare ones in two, three or four.

how long is a character

a code that does not work



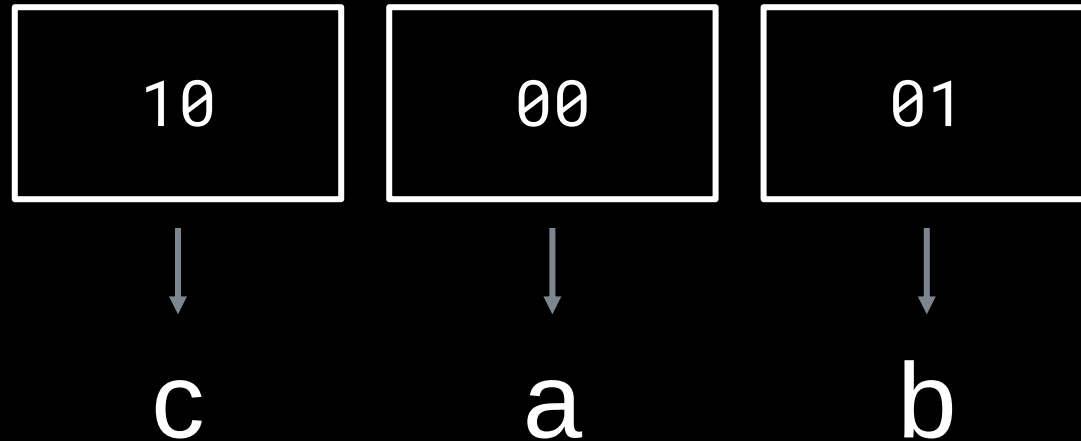
reading 1 0 · 10 → a c

reading 2 01 · 0 → b a

same bits, two readings. the receiver cannot know where a character ends.

fixed length

a = 00 b = 01 c = 10 d = 11



every character is two symbols. counting is enough, no separators needed.

fixed length is easy to read. variable length can be shorter.

how many symbols per letter?

your alphabet



4 colours = 2 bits per symbol

1 symbol per letter	4	4 letters	not enough
2 symbols per letter	4 x 4	16 letters	still not enough
3 symbols per letter	4 x 4 x 4	64 letters	enough for 26

26 letters into an alphabet of four: three symbols each, and 38 spare codes.

the same calculation, in general

$$N = k^n$$

k = symbols in your alphabet

n = symbols per character

N = characters you can code

the other way round:

$$n = \lceil \log_k(N) \rceil$$

examples

$$k = 4, n = 3$$

your colours, three per letter

$$N = 64$$

$$k = 2, n = 8$$

one byte

$$N = 256$$

$$k = 2, n = 1$$

a single light switch

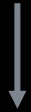
$$N = 2$$

26 letters, 4 colours: three symbols each, and always round up.

the same bits, other meanings

one byte sequence, three readings

48 65 6C 6C 6F



as ascii

"Hello"



as grey values

5 pixels



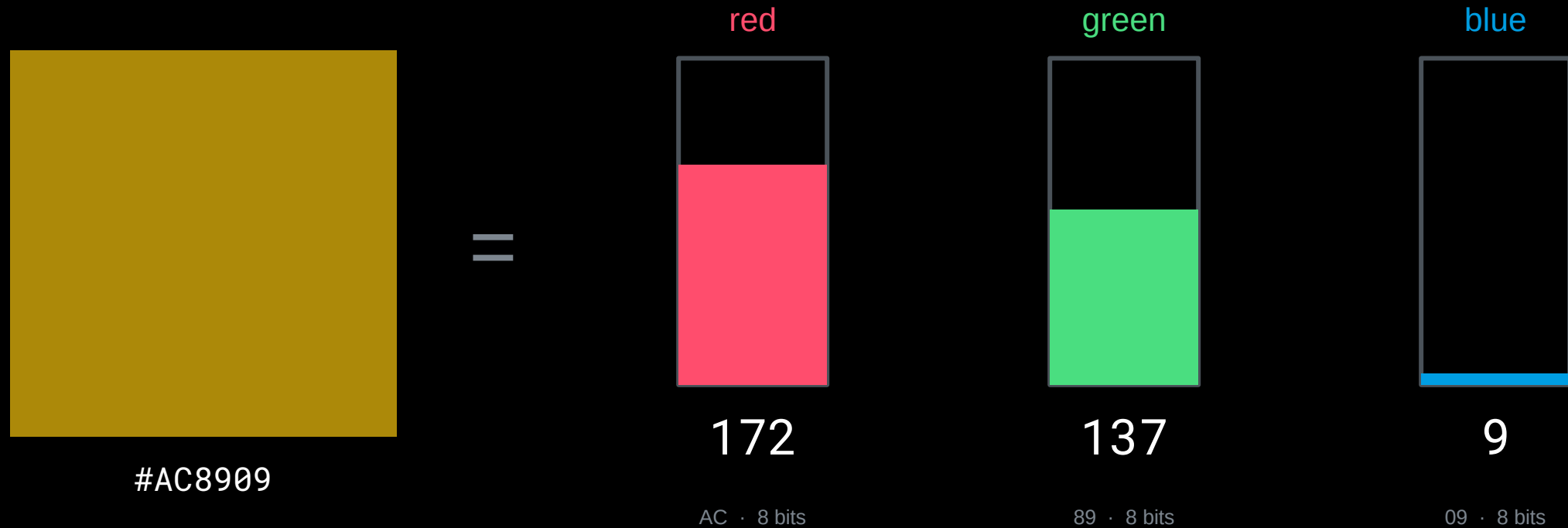
as audio

5 samples

the bytes do not say which one is right.

the type is another agreement: file extension, header,
or a type field in your own frame.

one pixel, three bytes



the same three numbers your sender writes: `led.set_color(172, 137, 9)`

$256 \times 256 \times 256 = 16.7$ million colours, 24 bits per pixel.

two ways to store a picture

bitmap

one colour per pixel

1920 x 1080 pixels

= 2 million values

zoom in and you see the pixels

paint one yourself:

winf-hsos.github.io/lifi-concept-demos

vector

instructions instead of pixels

circle: centre, radius, colour

= a few dozen values

redrawn at any size, always sharp

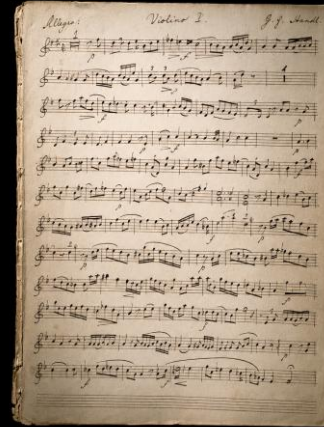
same picture, two agreements.
neither is right, they are good at different things.

code systems outside the computer



barcode

bar widths → digits



sheet music

position → pitch



braille

six dots → a character



dna

three bases → an amino acid

none of them needs a computer. all of them need an agreement.

today

- 1 agree on a code: letters to colour symbols
- 2 write it down, both directions
- 3 fill `send_text()` and `receive_text()`
- 4 one ten-letter word, hand-timed, fully correct